Fiscal Year:	FY 2020	**Task Last Updated:**	FY 04/09/2020
PI Name:	Goree, John Ph.D.		
Project Title:	Nonlinear Wave Experiments in Dusty Plasmas		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:	National Science Foundation	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	06/14/2017	**End Date:**	06/13/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Change in start and end dates to 6/14/2017-6/13/2021 (from 4/15/2017-4/15/2021) per I. Hahn/JPL and PI (Ed., 1/19/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Liu, Bin Ph.D. (University of Iowa)		
Grant/Contract No.:	JPL 1573629		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

A flight experiment will be performed and analyzed in a study of nonlinear synchronization of waves. Synchronization is a cooperative phenomenon for oscillators or waves. A signature of synchronization is a frequency that shifts when an external modulation is applied. Compared to point-like oscillators, waves are more complicated due to their spatio-temporal behavior and dispersion properties. An understanding of synchronization is lacking for waves of all kinds, including those in dusty plasmas.
The experiment will use dusty plasmas, also called "complex plasmas." These are mixtures of small particles of solid matter, electrons, ions, and neutral gas. The small solid particles obtain large electric charges, so that they exhibit collective behavior. They are electrically confined, so that they do not contact any boundaries.

Microgravity enables this experiment by suppressing sedimentation, thereby allowing the waves to propagate distances of many wavelengths, with relatively uniform background conditions. This experiment is based on previous ground-based experiments, where waves propagated only a few wavelengths.

Our flight experiment will be performed using “Plasma Kristall 4” (PK-4), which is a facility in the Columbus Module of the International Space Station (ISS), built by the European Space Agency (ESA).

SCIENTIFIC/TECHNICAL MERITS

The main scientific topic is the synchronization of a dust acoustic wave (DAW). A DAW is a low-frequency compressional wave, which involves charge-separation from electrons and ions. The DAW is self-excited by an ion flow, and it grows naturally to a large amplitude so that nonlinear effects occur readily. Synchronization experiments are performed by applying electrical waveforms to an electrode to modulate the dusty plasma sinusoidally. The diagnostic for detecting the wave motion is video imaging using particle observation cameras. Image data will be analyzed to prepare space-time diagrams, which provide a spatial and temporal description of the wave's amplitude as it propagates. These space-time diagrams are the foundation for further analysis methods.

In support of our experiment, a theoretical effort will be undertaken at Iowa. The goal is to contribute to the development of a theory for synchronization of propagating waves, to compare to the experiment. The theoretical effort at Iowa will focus on numerical simulations using a hybrid model. Molecular dynamics simulation will also be performed. The main theoretical effort, which will center on analytic theory, will be done by theorist elsewhere, and we will collaborate with them.

The feasibility of our flight experiment is based on our previous ground-based experiments, which confirm that synchronization of the DAW occurs. These ground-based experiments also revealed unexplained phenomena, including a nonharmonic synchronization state, which we will investigate with our flight data.

In the unlikely event that our main experiment with waves does not yield useful results, we have a plan for a backup experiment to investigate mobility-limited motion of particles having a different size.

BROADER IMPACTS

This research has an overlap with nonlinear dynamics as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, a website, and Twitter. A K-12 outreach presentation will be made at an elementary school.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Our accomplishment in this report period includes a new flight experiment and an advance in our understanding of nonlinear waves in three-dimensional dusty plasmas under microgravity. We planned and executed our second flight experiment on the International Space Station (ISS) in PK-4 Science Campaign 9, investigating the synchronization of nonlinear waves. We analyzed the flight data from our first flight experiment in Campaign 5. We made progress in our understanding of nonlinear dust acoustic waves (DAWs), which are low-frequency compressional waves self-excited in a collection of micron-size particles immersed in a neon plasma. These small particles obtained large electric charges by absorbing electrons and ions in the plasma.
ISS FLIGHT EXPERIMENT

Our second ISS flight experiment on nonlinear waves was successfully completed in PK-4 Science Campaign 9, February 2020. This flight experiment investigated nonlinear waves in three-dimensional dusty plasmas under microgravity conditions. It successfully accomplished its goals:

• implementation of a new scheme for varying the modulation amplitude for the synchronization of the waves.

• obtaining publishable data for a wide range of experiment parameters.

This new experiment was a major improvement, as compared with our first experiment in Campaign 5 on the ISS. First, we implemented our experiment in two time slots, doubling the experiment time. Second, we improved our experiment script by adding an option to vary modulation amplitude automatically.

One conclusion from this experiment is that another ISS flight experiment is needed for our wave synchronization project. For the past two ISS experiments, we have successfully identified the conditions for wave excitation, and we have also implemented the scheme for wave modulation. However, for better synchronization we need to fine tune gas pressure to obtain more stable waves than what we had in last two experiments.

Co-Investigator (CoI) Bin Liu participated in the execution of this experiment in Toulouse, France, together with PK-4 Core-Team scientists. He also observed other PK-4 flight experiments performed in Campaign 9. To prepare for this flight experiment, we worked closely with PK-4 Core Team scientists. At Iowa, we designed our experiment procedure, configuration, and parameters regime, yielding a flow chart. PK-4 Core Team generated an experiment script based on our flow chart. Co-I Bin Liu travelled to the DLR (German Space Agency), and tested the script using the PK-4 ground model at the DLR.

ANALYSIS OF ISS PK-4 FLIGHT DATA

Task Progress:	During this report period, we analyzed the flight data from our first ISS PK-4 flight experiment. Major results include: the spectrum and the dispersion relation for self-excited dust acoustic waves; the spatial growth of waves; density correlation functions; transverse wave motion and vortices; indication of partial synchronization and frequency clustering, identification of synchronization states, and nonlinear interactions. Wave spectrum and dispersion relation. Using particle position, we calculated microscopic particle density, yielding a wave spectrum of self-excited dust acoustic waves in a dusty plasma under microgravity condition. We found that the wave spectrum is broadband, meaning that an ensemble of waves was excited. Based on the wave spectrum, we also obtained a dispersion relation, which surprisingly coincides with a dispersion relation predicted by a linear theory. Spatial growth of waves. Using flow velocity, we discovered different stages of the growth of the DAWs propagating across the dust cloud. We found that waves grew exponentially as soon as they were excited near the center of the dust cloud. As the waves approached the edge of the cloud the growth slowed down significantly. The flow velocity was obtained using particle image velocimetry (PIV); the PK-4 group at Auburn University assisted us in obtaining the flow velocity data. Density correlation functions. We computed the correlation function of microscopic particle density, finding that the correlation function exhibited a damped oscillation. We fitted the correlation function to an expression that is for a dusty plasma crystal, yielding a measurement of wave frequency and damping rate. We found that the wave damping rate varies among different modes and the most unstable modes had the smallest damping rate, as expected. Transverse wave motion and vortices. We observed a transverse wave dispersion relation, which is not expected for our dusty plasma near non-ideal gas regime. We attribute this surprising result to vortices motion of waves. The vorticity was calculated using the flow velocity data. Indication of partial synchronization and frequency clustering. We discovered a new topic that we can explore using our flight experiment. We found an indication of spatial frequency clustering, which is a partial synchronization in some regions of the dust cloud. In our most recent flight experiment in February 2020, we have included this topic in our experiment script. Our dusty plasma can be used as a model for a chain of nonlinear coupled oscillators. Identification of synchronization states. We calculated power spectra for dust acoustic waves under external modulation, finding indications of synchronization states. However, a better frequency resolution is required to identify the synchronization more accurately. We have improved on this in our most recent flight experiment on the ISS. Bispectrum and nonlinear interactions. To understand nonlinear phenomena such as frequency clustering and synchronization in our dusty plasma, we calculated the bispectrum of particle motion. Although we have found some indication of three-wave mixing, more thorough analysis is needed to determine which nonlinear processes dominate in our plasma system, for example, three-wave mixing or wavefront merging. PK-4 FLIGHT DATA During this report period, the Iowa group received a portion of the flight data from Science Campaigns 7, which was from our flight experiment for pulsed shear motion. FLIGHT EXPERIMENT PLANNING Our planning involved a collaboration with other institutions or agencies. Significant collaborators included PK-4 Core Team members in Germany and Russia, PK-4 Facility Science Team (FST), ESA (European Space Agency) officials, CADMOS (Microgravity and Spatial Operations Development Support Center) at the CNES (France National Center for Space Studies), and the collaborators specific to Iowa projects. Both PI John Goree and Co-I Bin Liu interacted with these collaborators frequently via email and occasionally by phone, and we also met them at various meetings and conferences. One significant meeting was the recent PK-4 meeting at the DLR (German Space Agency) in December 2019. Co-I Bin Liu attended this meeting, and he gave a talk regarding the progress of Iowa PK-4 projects. THEORY During this period, theory was developed, together with collaborators in India (Prof. S. Tiwari, and Prof. Abhijit Sen) and Naval Research Laboratory (Dr. Guru Ganguli, and Dr. C. Crabtree). The theory is nonlinear wave mixing, under conditions related to our experiments.
Bibliography Type:	Description: (Last Updated: 01/23/2025)
Abstracts for Journals and Proceedings	Goree J, Liu B. "Dusty plasma experiments under microgravity conditions." Presented at the 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Mir AA, Tiwari S, Sen A, Ganguli G, Crabtree C, Liu B, Goree J. "Beat waves in dusty plasma." Presented at the 12th International Conference on Plasma Science and Applications (ICPSA), Lucknow, India, November 11-14, 2019. Abstracts. 12th International Conference on Plasma Science and Applications (ICPSA), Lucknow, India, November 11-14, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Liu B, Goree J, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Usachev AD, Molotkov VI, Petrov OF, Thoma MH. "Nonlinear dust acoustic waves in a plasma under microgravity conditions." Presented at the 61st Annual Meeting of the American Physical Society Division of Plasma Physics, Fort Lauderdale, Florida, October 21-25, 2019. Bulletin of the American Physical Society. 2019;64(11):NP10.00075. http://meetings.aps.org/Meeting/DPP19/Session/NP10.75 , Oct-2019
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